

AI Use Case: “Rashid” - Spatial Analysis Service (demo)



Use Case details for technical feasibility & commercial impact assessment

ENGAGEMENT CONTEXT



Engagement Manager

Gaafar Al-Marzouki



Client Entity

Ministry of Municipalities and Housing



Engagement Name

Urban Planning RRT – Pipeline Project



Approx. Date Required

Q3 2026

Use Case Type: ☐ Internal Product (for EMs) ☒ Client Deliverable ☐ Both

USE CASE DETAILS



Problem Statement / Pain Point

Business-led spatial analysis requests currently require substantial manual effort from the GIS department. Teams must interpret the request, identify relevant datasets, retrieve them from the SDI, compare multiple years of spatial data, perform the analysis, and prepare outputs. This process is time-consuming, difficult to scale, and may lead to inconsistent analytical approaches across cities and requests. Rashid is already positioned around spatial analysis as one of its core operating domains, but the current general use case needs to be narrowed to this specific service.



Desired Outcome / Success Criteria

Enable Rashid to handle spatial analysis requests end-to-end within the GIS department by receiving a business request, identifying and retrieving the required datasets from the SDI, performing the analysis, and producing clear spatial and statistical outputs. For this demo, Rashid should identify urban expansion in the major cities of the Kingdom over one year by comparing current and previous-year city boundaries and quantifying the increase in built-up urban extent.

IMPACT, DATA & TIMELINE



Priority

☐ Critical ☒ High ☐ Medium ☐ Low



Expected PoC Timeline

☒ < 6 Wks ☐ 6-8 Wks ☐ 2-4 Mo ☐ 4-6 Mo ☐ 6+ Mo



Estimated Impact

Hours saved / month: Reduced analyst effort in recurring multi-city spatial analysis requests



Data / Systems Available

ArcGIS Enterprise, SDI / Spatial Data Infrastructure, satellite imagery, subdivision plans, urban boundary / development boundary layers, land use layers, building layers, knowledge portal



Dependencies / Blockers

Availability of comparable current and previous-year datasets; access to SDI and imagery sources; agreed methodology for measuring urban expansion; ArcGIS Enterprise integration; human review and validation of final outputs where needed

Request Example

A request is received to **identify urban expansion in the major cities of the Kingdom over one year**. i.e., how much each city expanded between the previous year and the current year.

Rashid Workflow

1. Receive and interpret the request from the business owner
2. Define the spatial and time scope of the analysis
3. Search the SDI for all relevant datasets
4. Retrieve the required inputs, such as satellite imagery, subdivision plans, urban boundary layers, land use data, and other supporting layers
5. Prepare comparable current and previous-year data inputs
6. Analyze the data and delineate city boundaries for both years
7. Identify expansion areas between the two periods
8. Generate outputs including maps, boundaries, and statistics

Expected Outputs

- Previous-year urban boundary
- Current-year urban boundary
- Expansion zones identified spatially
- Statistics by city, such as:
 - previous urban area
 - current urban area
 - expansion area in sq. km
 - growth percentage